

Swiggy Sales Analysis

Problem Statement

Swiggy lacks a centralized view of sales performance across states, cities, and time periods, making it difficult to identify high-growth regions, underperforming quarters, and popular food categories. Manual tracking of 20,000+ orders from multiple restaurants leads to missed revenue opportunities and inefficient inventory planning. This project builds an interactive dashboard and Python-based analysis to uncover sales trends, top cities/restaurants, and category-wise revenue contributions for data-driven decision-making.

Import Libraries

```
[151]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

Import Data

```
[152]: df = pd.read_csv("swiggy_data.csv")
```

```
[153]: df["Order Date"] = pd.to_datetime(df["Order Date"], errors="coerce")
```

```
[154]: df.head(5)
```

[154]:		State	City	Order Date	Restaurant Name	Location	Category	Dish Name	Price (INR)	Rating	Rating Count
	0	Karnataka	Bengaluru	2025-06-29	Anand Sweets & Savouries	Rajarajeshwari Nagar	Snack	Butter Murukku-200gm	133.9	4.0	0
	1	Karnataka	Bengaluru	2025-04-03	Srinidhi Sagar Deluxe	Kengeri	Recommended	Badam Milk	52.0	4.5	25
	2	Karnataka	Bengaluru	2025-01-15	Srinidhi Sagar Deluxe	Kengeri	Recommended	Chow Chow Bath	117.0	4.7	48
	3	Karnataka	Bengaluru	2025-04-17	Srinidhi Sagar Deluxe	Kengeri	Recommended	Kesari Bath	65.0	4.6	65
	4	Karnataka	Bengaluru	2025-03-13	Srinidhi Sagar Deluxe	Kengeri	Recommended	Mix Raitha	130.0	4.0	0

```
[155]: df.tail(5)
```

[155]:		State	City	Order Date	Restaurant Name	Location	Category	Dish Name	Price (INR)	Rating	Rating Count
	197425	Sikkim	Gangtok	2025-01-25	Mama's Kitchen	Gangtok	Momos	Soya cheese chilli momo	112.0	4.4	0
	197426	Sikkim	Gangtok	2025-07-02	Mama's Kitchen	Gangtok	Momos	Kurkure momo fried	140.0	4.4	0
	197427	Sikkim	Gangtok	2025-03-25	Mama's Kitchen	Gangtok	Momos	Chilli cheese momo	126.0	4.4	0
	197428	Sikkim	Gangtok	2025-03-26	Mama's Kitchen	Gangtok	Momos	Veg Momos (8 Pc)	85.0	4.4	0
	197429	Sikkim	Gangtok	2025-03-27	Mama's Kitchen	Gangtok	Momos	Soya Momo	100.0	4.4	0

Data Exploration and Validation

```
[156]: print("No of Rows:", df.shape[0])
```

No of Rows: 197430

```
[157]: print("No of Fields:", df.shape[1])
```

No of Fields: 10

```
[178]: df.columns
```

```
[178]: Index(['State', 'City', 'Order Date', 'Restaurant Name', 'Location',  
         'Category', 'Dish Name', 'Price (INR)', 'Rating', 'Rating Count',  
         'YearMonth', 'DayName', 'Food Category', 'Quarter'],  
        dtype='str')
```

```
[179]: df.duplicated().sum()
```

```
[179]: np.int64(0)
```

```
[181]: df = df.drop_duplicates()
```

```
[158]: df.info()
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 197430 entries, 0 to 197429  
Data columns (total 10 columns):  
#   Column                Non-Null Count  Dtype  
---  ---  
0   State                 197430 non-null  str  
1   City                 197430 non-null  str  
2   Order Date           197430 non-null  datetime64[us]  
3   Restaurant Name       197430 non-null  str
```

```
[159]: df.isnull().sum()
```

```
[159]: State      0
City      0
Order Date  0
Restaurant Name  0
Location   0
Category   0
Dish Name  0
Price (INR)  0
Rating     0
Rating Count  0
dtype: int64
```

```
[182]: df.describe()
```

```
[182]:
```

	Order Date	Price (INR)	Rating	Rating Count
count	197403	197403.000000	197403.000000	197403.000000
mean	2025-05-01 19:42:28.028145	268.501413	4.341579	28.323637
min	2025-01-01 00:00:00	0.950000	1.500000	0.000000
25%	2025-03-01 00:00:00	139.000000	4.300000	0.000000
50%	2025-05-02 00:00:00	229.000000	4.400000	2.000000
75%	2025-07-01 00:00:00	329.000000	4.500000	15.000000
max	2025-08-31 00:00:00	8000.000000	5.000000	999.000000
std	NaN	219.222980	0.422585	87.547541

Charts Design

Monthly Revenue Trend

```
[167]: df["Order Date"] = pd.to_datetime(df["Order Date"])

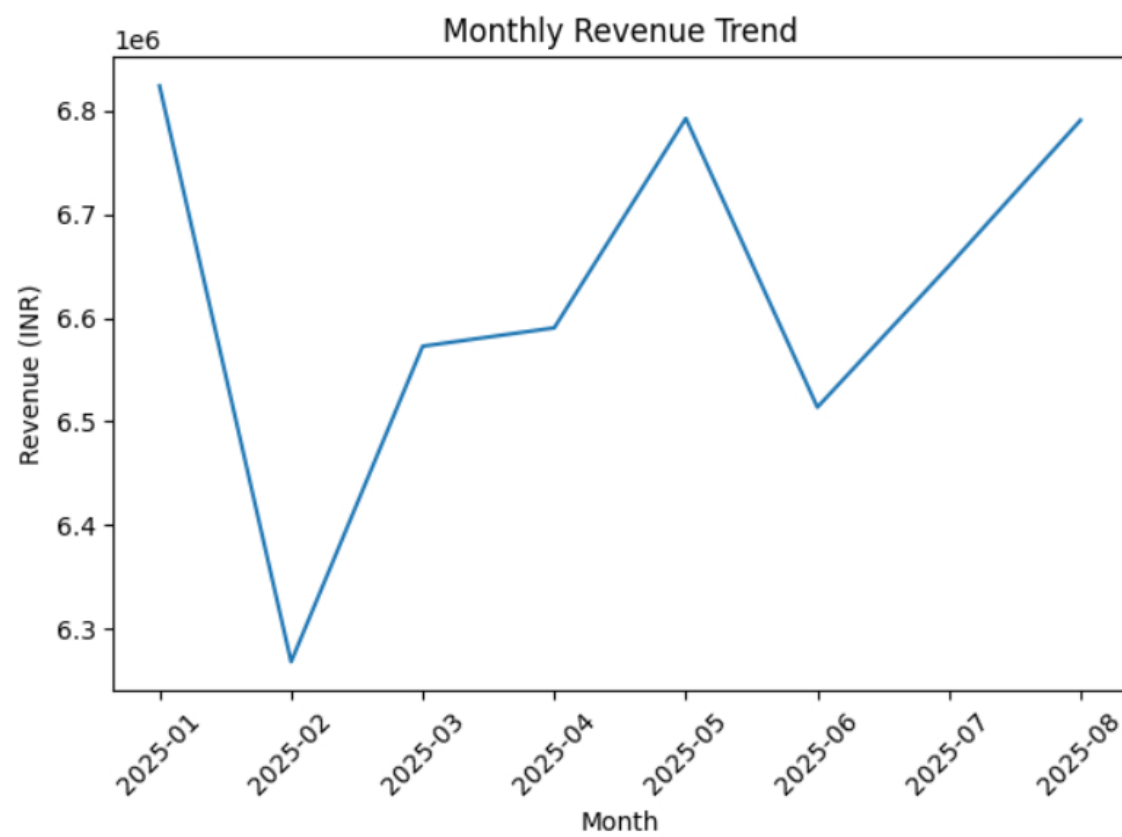
df["YearMonth"] = df["Order Date"].dt.to_period("M").astype(str)

monthly_revenue = df.groupby("YearMonth")["Price (INR)"].sum().reset_index()

plt.figure()
plt.plot(monthly_revenue["YearMonth"], monthly_revenue["Price (INR)"])

plt.xticks(rotation=45)
plt.xlabel("Month")
plt.ylabel("Revenue (INR)")
plt.title("Monthly Revenue Trend")

plt.tight_layout()
plt.show()
```



Daily Revenue Trend

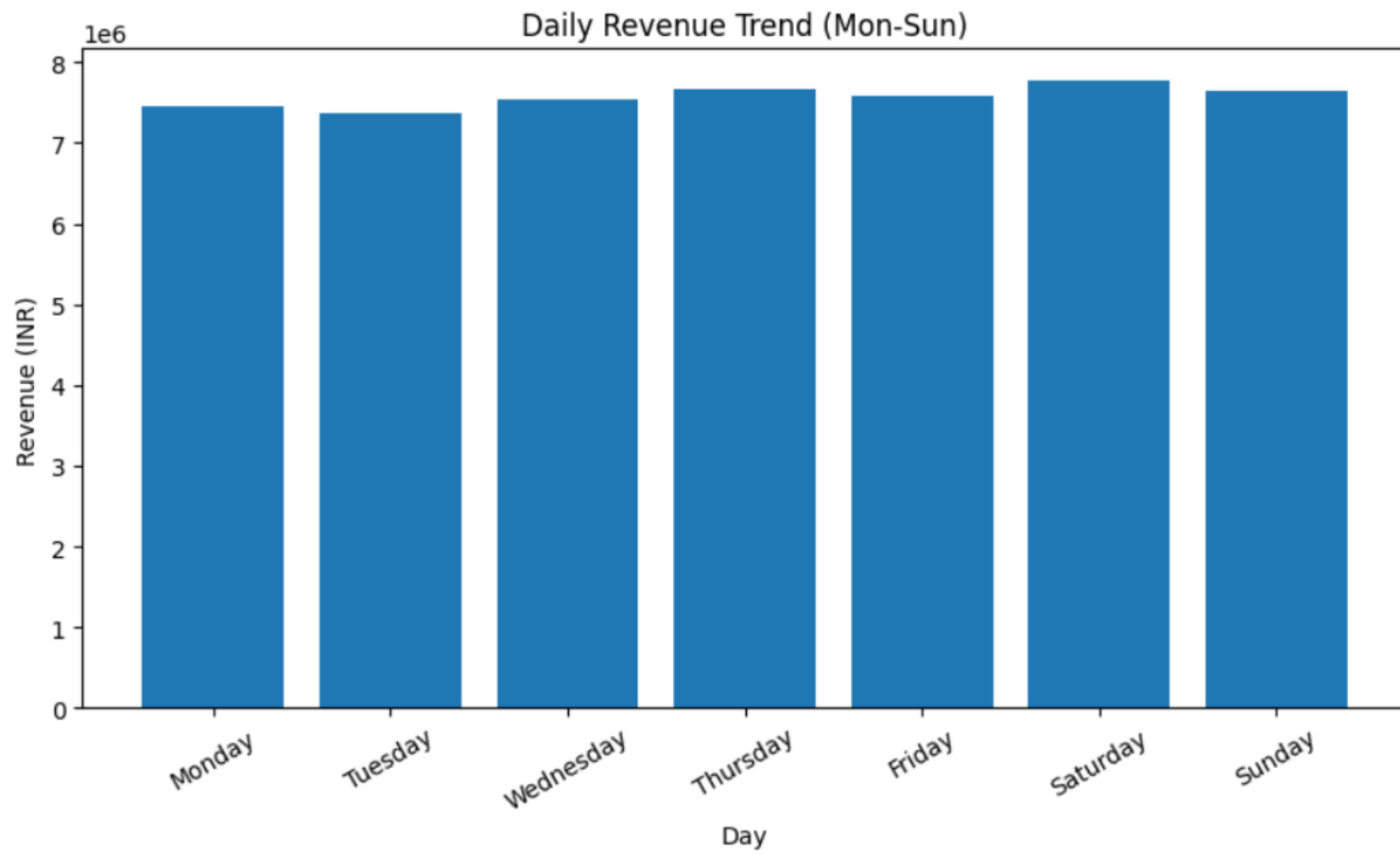
```
[168]: df["DayName"] = pd.to_datetime(df["Order Date"]).dt.day_name()

daily_revenue = (
    df.groupby("DayName")["Price (INR)"]
      .sum()
      .reindex([
          "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"
      ])
)

plt.figure(figsize=(10, 5))
plt.bar(daily_revenue.index, daily_revenue.values)

plt.title("Daily Revenue Trend (Mon-Sun)")
plt.xlabel("Day")
plt.ylabel("Revenue (INR)")
plt.xticks(rotation=30)

plt.show()
```



Total Sales by Food Type (Veg vs Non-Veg)

[169]: *# Define Non-Veg keywords*

```
non_veg_keywords = [  
    "chicken", "egg", "fish", "mutton",  
    "prawn", "biryani", "kabab", "kebab",  
    "non-veg", "non veg"  
]
```

[170]: *# Create Food Category column (corrected regex join)*

```
df["Food Category"] = np.where(  
    df["Dish Name"].str.lower().str.contains("|".join(non_veg_keywords), na=False),  
    "Non-Veg",  
    "Veg"  
)
```

[171]: *# Aggregate revenue by category*

```
food_revenue = (  
    df.groupby("Food Category")["Price (INR)"]  
    .sum()  
    .reset_index()  
)
```

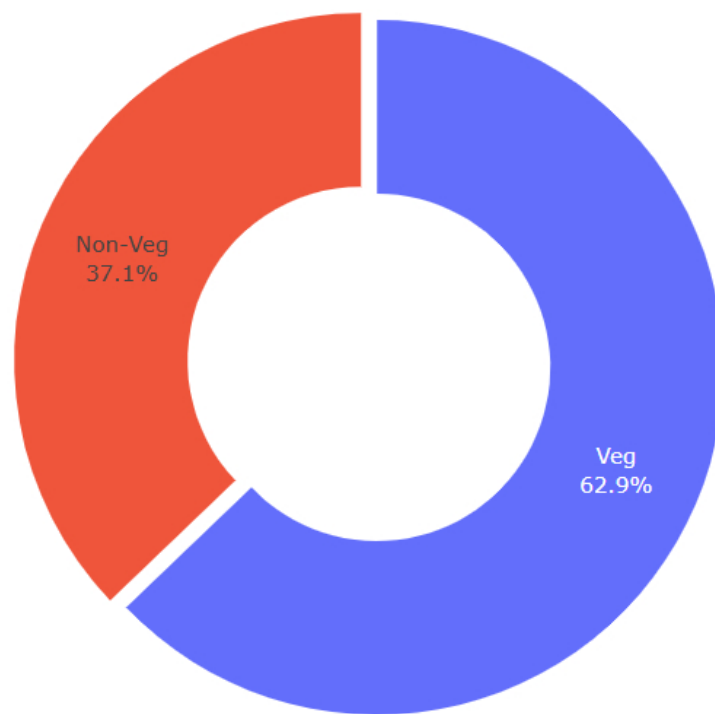
```
[172]: # Create donut pie chart
fig = px.pie(
    food_revenue,
    values="Price (INR)",
    names="Food Category",
    hole=0.5,
    title="Revenue Contribution: Veg vs Non-Veg",
)

fig.update_traces(
    textinfo="percent+label",
    pull=[0.05, 0] # Slightly pull the first slice (Veg/Non-Veg)
)

fig.update_layout(
    height=500,
    margin=dict(t=60, b=40, l=40, r=40)
)

fig.show()
```

Revenue Contribution: Veg vs Non-Veg

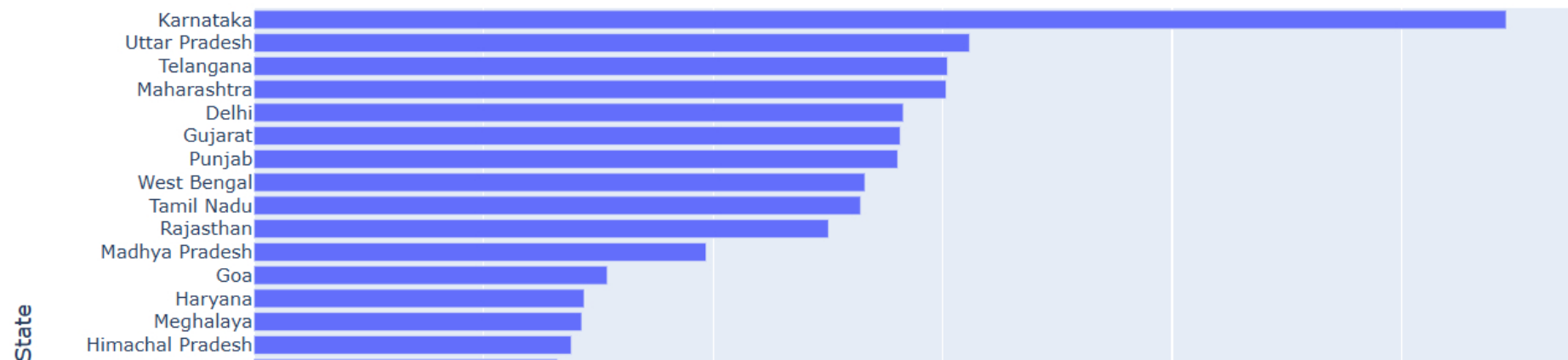


Veg
Non-Veg

Total Sales by State

```
[173]: fig = px.bar(  
    df.groupby("State", as_index=False)["Price (INR)"].sum()  
    .sort_values("Price (INR)", ascending=False),  
    x="Price (INR)",  
    y="State",  
    orientation="h",  
    title="Revenue by State (INR)"  
)  
  
fig.update_layout(height=600, yaxis=dict(autorange="reversed"))
```

Revenue by State (INR)



Quarterly Performance Summary

```
[174]: df["Quarter"] = df["Order Date"].dt.to_period("Q").astype(str)

quarterly_summary = (
    df.groupby("Quarter", as_index=False)
    .agg(
        Total_Sales=("Price (INR)", "sum"),
        Avg_Rating=("Rating", "mean"),
        Total_Orders=("Order Date", "count")
    )
    .sort_values("Quarter")
)

quarterly_summary["Total_Sales"] = quarterly_summary["Total_Sales"].round(0)
quarterly_summary["Avg_Rating"] = quarterly_summary["Avg_Rating"].round(2)

quarterly_summary
```

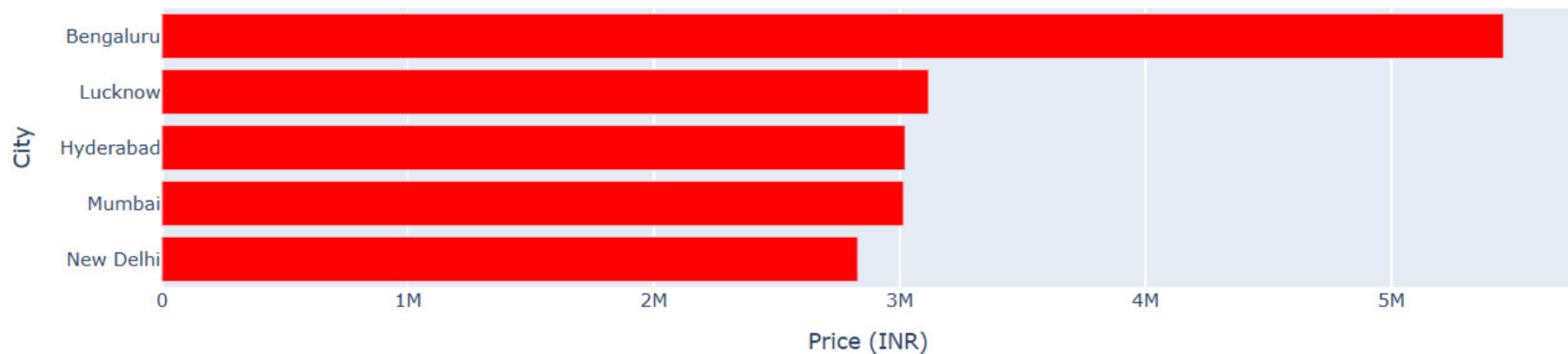
[174]:

	Quarter	Total_Sales	Avg_Rating	Total_Orders
0	2025Q1	19664881.0	4.34	73085
1	2025Q2	19896936.0	4.34	74155
2	2025Q3	13441168.0	4.34	50163

Top 5 Cities by Sales

```
[175]: top_5_cities = (  
    df.groupby("City")["Price (INR)"]  
    .sum()  
    .nlargest(5)  
    .sort_values()  
    .reset_index()  
)  
  
fig = px.bar(  
    top_5_cities,  
    x="Price (INR)",  
    y="City",  
    orientation="h",  
    title="Top 5 Cities by Sales (INR)",  
    color_discrete_sequence=["red"]  
)  
  
fig.show()
```

Top 5 Cities by Sales (INR)



Key Takeaways

- Total Sales: ₹53M | Total Orders: 197K | Avg Order Value: ₹268 | Avg Rating: 4.3
- Top 5 States by revenue: Karnataka, Uttar Pradesh, Telangana, Maharashtra, Delhi
- Top 5 Cities: Bengaluru, Hyderabad, Ahmedabad, Chennai, Lucknow (New Delhi & Kolkata in top 10)
- Q3 sales dropped 32% vs Q2 (₹19.9M → ₹13.4M) – critical decline
- Weekend revenue (Sat+Sun) is 34% higher than weekdays
- Non-Veg contributes 68% of total revenue (Veg 32%)
- Top restaurants by order volume: McDonald's, KFC, Burger King, Pizza Hut, Domino's
- Best-selling dishes have balanced contribution (16-23% each)
- Low-revenue states: Goa, Meghalaya, Himachal, J&K – expansion opportunity

▼ Conclusion

The Swiggy sales analysis shows heavy revenue concentration in top 5 states and cities, with Karnataka leading. Despite a strong average rating of 4.3, the 32% sales decline in Q3 2025 is a major red flag requiring root-cause analysis (competitor offers, delivery issues, or seasonal dip). Non-Veg categories and fast-food chains (McDonald's, KFC) dominate revenue, so menu and partnership focus should remain there. Weekend spikes suggest targeted promotions from Friday to Sunday would maximize ROI. The Power BI dashboard + Python visualizations enable real-time monitoring of state-wise performance, quarterly trends, and restaurant-level strategies.